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CUET UG

Mathematics

Vectors and Three-Dimensional Geometry

Vector Algebra, Products, Direction Cosines, 3D Lines & Shortest Distance · CUET UG Section B1 Unit IV

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus & Exam Weightage

In the NTA CUET (UG) Mathematics examination (Section B1, Unit IV), **Vectors and Three-Dimensional Geometry** constitutes one of the most scoring and computationally dense sections of the paper. Typically accounting for **6 to 8 questions** (30 to 40 marks out of 200), this unit assesses algebraic mastery of geometric entities in space.

Sub-Topic	Official NTA Syllabus Coverage	High-Yield CUET Focus
Vector Algebra & Components	Scalars, vectors, magnitude, direction cosines & ratios of vectors; unit, equal, zero, collinear & coplanar vectors; position vector, section formula.	Section formula (internal & external), unit vector along angle bisector, collinearity checks.
Scalar (Dot) Product	Definition, geometric interpretation, properties, projection of a vector on another line/vector, work done, orthogonality condition.	Orthogonal projection scalar/vector, Cauchy-Schwarz, angle between vectors.
Vector (Cross) Product	Definition, determinant formulation, unit normal vector, area of triangle and parallelogram, Lagrange's identity.	Area of triangle/parallelogram from diagonals vs sides, Lagrange identity shortcuts.
Scalar Triple Product (STP)	Definition $[(a)\ (b)\ (c)] = (a)\cdot((b)\times(c))$, determinant evaluation, coplanarity of 3 vectors, volume of parallelepiped & tetrahedron.	Volume of parallelepiped, condition of coplanarity with unknown parameter λ.
Direction Cosines & Ratios	Direction angles α, β, γ, relation $l^2+m^2+n^2=1$, direction ratios (a,b,c), DCs of line joining two points.	Identity $\sin^2\alpha+\sin^2\beta+\sin^2\gamma=2$, converting non-standard line forms.
Equations of Lines in Space	Vector and Cartesian forms through a point with given direction, through two points; angle between two lines; collinearity of points.	Standard form conversion $\frac{2x-1}{4}=\frac{1-y}{3}$, intersection of lines, foot of perpendicular.
Shortest Distance (SD)	Skew lines vs parallel lines; vector and Cartesian formulas for shortest distance; condition of intersection of two lines in space.	Determinant formula for SD, verifying intersecting lines (SD = 0), distance between parallel lines.

CUET EXAM ALERT — THE NON-STANDARD FORM TRAP

CUET test designers routinely disguise direction ratios by giving line equations where the coefficients of x, y, z are not $+1$. For instance, in $\frac{2x-3}{4}=\frac{5-y}{2}=\frac{3z+1}{6}$, students who pick $(4,2,6)$ as direction ratios immediately score negative marks. You must normalize each numerator to have a $+1$ leading coefficient:

$$\frac{x-\frac{3}{2}}{2}=\frac{y-5}{-2}=\frac{z-\left(-\frac{1}{3}\right)}{2} \implies \text{Direction ratios are } (2,-2,2) \text{ or } (1,-1,1).$$

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Vector Algebra Fundamentals

Definition, Magnitude & Basic Types

A **vector** is a directed line segment characterized by both magnitude (length) and spatial orientation (direction). If a point $A(x_1, y_1, z_1)$ is the initial point and $B(x_2, y_2, z_2)$ is the terminal point, the directed line segment $\overrightarrow{AB}$ represents a vector:

$$\overrightarrow{AB} = (x_2 - x_1)\hat{\imath} + (y_2 - y_1)\hat{\jmath} + (z_2 - z_1)\hat{k}$$

Its **magnitude** (length or norm), denoted $|\overrightarrow{AB}|$, is the Euclidean distance:

$$|\overrightarrow{AB}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

CLASSIFICATION OF VECTOR TYPES

- **Zero (Null) Vector ((0)):** Magnitude is 0; initial and terminal points coincide. Its direction is indeterminate (or arbitrary).
- **Unit Vector ($\hat{a}$):** A vector having unit magnitude ($|\hat{a}| = 1$). For any non-zero vector (a), its unit vector is:

$$\hat{a} = \frac{(a)}{|(a)|}$$

- **Collinear (Parallel) Vectors:** Vectors that act along the same straight line or parallel straight lines, regardless of magnitude or sense. Two vectors (a) and (b) are collinear if and only if there exists a scalar $\lambda \neq 0$ such that:

$$(b) = \lambda(a) \iff \frac{b_1}{a_1} = \frac{b_2}{a_2} = \frac{b_3}{a_3} = \lambda$$

- **Co-initial Vectors:** Two or more vectors possessing the exact same initial starting point.
- **Coplanar Vectors:** Three or more vectors whose line supports are parallel to the same plane.
- **Equal Vectors:** Vectors possessing identical magnitude and identical direction, regardless of their position in space.
- **Negative of a Vector ($-(a)$):** Same magnitude as (a), but exactly opposite direction ($\overrightarrow{BA} = -\overrightarrow{AB}$).

Position Vector and Section Formula

Let $O(0, 0, 0)$ be the fixed origin of reference in three-dimensional space. The **position vector** of any point $P(x, y, z)$ is:

$$(r) = \overrightarrow{OP} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$$

For two distinct points A and B with position vectors (a) and (b) respectively:

Configuration	Position Vector of Point P
Internal Division $(m : n)$	$(r) = \frac{m(b)+n(a)}{m+n}$
External Division $(m : n)$	$(r) = \frac{m(b)-n(a)}{m-n} \quad (m \neq n)$
Mid-point $(1 : 1)$	$(r) = \frac{(a)+(b)}{2}$
Centroid of Triangle ABC	$(r)_G = \frac{(a)+(b)+(c)}{3}$

Centroid of Tetrahedron $ABCD$	$(r)_G = \frac{(a)+(b)+(c)+(d)}{4}$

INTERNAL ANGLE BISECTOR VECTOR FORMULA

If (a) and (b) are two non-collinear co-initial vectors defining an angle θ, a vector directed along the **internal angle bisector** is given by:

$$(v)_{\text{bisector}} = \lambda\left(\hat{a}+\hat{b}\right) = \lambda\left(\frac{(a)}{|(a)|}+\frac{(b)}{|(b)|}\right)$$

Similarly, a vector directed along the **external angle bisector** is $(v)_{\text{ext}} = \mu\left(\hat{a}-\hat{b}\right)$.

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Products of Vectors

The Scalar (Dot) Product

The **scalar product** (or dot product) of two vectors (a) and (b) separated by an angle θ $(0 \le \theta \le \pi)$ is a scalar quantity defined by:

$$(a)\cdot(b) = |(a)|\,|(b)|\cos\theta$$

Fundamental Property	Mathematical Formulation & Interpretation
Commutativity	$(a)\cdot(b) = (b)\cdot(a)$
Distributivity over Addition	$(a)\cdot((b)+(c)) = (a)\cdot(b)+(a)\cdot(c)$
Cartesian Component Form	$(a)\cdot(b) = a_1b_1+a_2b_2+a_3b_3$ where $(a) = a_1\hat{\imath}+a_2\hat{\jmath}+a_3\hat{k}$
Self-Product (Magnitude Squared)	$(a)\cdot(a) = \lvert(a)\rvert^2 \Longrightarrow \lvert(a)\rvert = \sqrt{(a)\cdot(a)}$
Orthogonality Criterion	$(a)\perp(b) \iff (a)\cdot(b) = 0$ (for non-zero vectors)
Angle Between Two Vectors	$\cos\theta = \frac{(a)\cdot(b)}{\lvert(a)\rvert\,\lvert(b)\rvert} = \frac{a_1b_1+a_2b_2+a_3b_3}{\sqrt{a_1^2+a_2^2+a_3^2}\sqrt{b_1^2+b_2^2+b_3^2}}$
Orthonormal Unit Vectors	$\hat{\imath}\cdot\hat{\imath} = \hat{\jmath}\cdot\hat{\jmath} = \hat{k}\cdot\hat{k} = 1$, and $\hat{\imath}\cdot\hat{\jmath} = \hat{\jmath}\cdot\hat{k} = \hat{k}\cdot\hat{\imath} = 0$

Projections of Vectors

Projection Type	Mathematical Formula
Scalar Projection of (a) on (b)	$p_s = (a)\cdot\hat{b} = \frac{(a)\cdot(b)}{\lvert(b)\rvert}$
Vector Projection of (a) on (b)	$(p)_v = \left(\frac{(a)\cdot(b)}{\lvert(b)\rvert}\right)\hat{b} = \left(\frac{(a)\cdot(b)}{\lvert(b)\rvert^2}\right)(b)$
Vector Component of $(a)\perp(b)$	$(a)_\perp = (a)-(p)_v = (a)-\left(\frac{(a)\cdot(b)}{\lvert(b)\rvert^2}\right)(b)$

CLASSIC CUET TRAPS IN DOT PRODUCT EXPANSION

1. **Scalar Projection Sign:** Scalar projection can be negative if $\theta > 90^\circ$. However, if asked for the "length of projection", take the absolute value $|(a)\cdot\hat{b}|$.
2. **Dot Product Cannot be Cancelled:** If $(a)\cdot(b) = (a)\cdot(c)$, it does **NOT** imply $(b) = (c)$. It merely implies $(a)\cdot((b)-(c)) = 0$, meaning (a) is perpendicular to $((b)-(c))$, or $(a) = (0)$, or $(b) = (c)$.

3. **Expanding Modulus Sums:** Never write $|(a)+(b)|^2 = a^2+b^2+2ab$. Always expand via self-dot product:

$$|(a)+(b)|^2 = ((a)+(b))\cdot((a)+(b)) = |(a)|^2+|(b)|^2+2((a)\cdot(b))$$

The Vector (Cross) Product

The **vector product** (or cross product) of two vectors (a) and (b) is a vector quantity perpendicular to the plane containing both vectors:

$$(a)\times(b) = (|(a)|\,|(b)|\sin\theta)\hat{n}$$

where θ $(0 \le \theta \le \pi)$ is the angle between them, and $\hat{n}$ is a unit vector perpendicular to both (a) and (b) whose sense is governed by the **right-hand thumb rule**.

Property	Mathematical Formula & Implication
Anti-Commutativity	$(b)\times(a) = -((a)\times(b))$
Collinearity Criterion	$(a)\times(b) = (0) \iff (a) \parallel (b)$ (for non-zero vectors)
Self-Cross Product	$(a)\times(a) = (0)$
Orthonormal Basis Cycle	$\hat{\imath}\times\hat{\jmath} = \hat{k}$, $\hat{\jmath}\times\hat{k} = \hat{\imath}$, $\hat{k}\times\hat{\imath} = \hat{\jmath}$; $\hat{\jmath}\times\hat{\imath} = -\hat{k}$, etc.
Cartesian Determinant Form	$(a)\times(b) = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3-a_3b_2)\hat{\imath}-(a_1b_3-a_3b_1)\hat{\jmath}+(a_1b_2-a_2b_1)\hat{k}$
Unit Vector Perpendicular to Plane	$\hat{n} = \pm\frac{(a)\times(b)}{\lvert(a)\times(b)\rvert}$
Sine of Angle Between Vectors	$\sin\theta = \frac{\lvert(a)\times(b)\rvert}{\lvert(a)\rvert\,\lvert(b)\rvert}$

Geometrical Applications & Identities

Geometric Figure	Area Formula
Parallelogram (Adjacent sides $(a),(b)$)	Area $= \lvert(a)\times(b)\rvert$
Parallelogram (Diagonals $(d)_1,(d)_2$)	Area $= \frac{1}{2}\lvert(d)_1\times(d)_2\rvert$
Triangle (Adjacent sides $(a),(b)$)	Area $= \frac{1}{2}\lvert(a)\times(b)\rvert$
Triangle with Vertices $A((a)),B((b)),C((c))$	Area $= \frac{1}{2}\lvert(a)\times(b)+(b)\times(c)+(c)\times(a)\rvert$

LAGRANGE'S IDENTITY — SUPREME CUET SPEED TOOL

For any two vectors (a) and (b):

$$|(a)\times(b)|^2 + ((a)\cdot(b))^2 = |(a)|^2\,|(b)|^2$$

In expanded form: $|(a)\times(b)|^2 = |(a)|^2\,|(b)|^2 - ((a)\cdot(b))^2 = \begin{vmatrix} (a)\cdot(a) & (a)\cdot(b) \\ (a)\cdot(b) & (b)\cdot(b) \end{vmatrix}$.

Direct Application: If $|(a)| = 5$, $|(b)| = 13$, and $(a)\cdot(b) = 25$, calculate $|(a)\times(b)|$ without finding θ:

$$|(a)\times(b)|^2 = (5^2)(13^2) - 25^2 = 25\times169 - 625 = 4225 - 625 = 3600 \implies |(a)\times(b)| = 60.$$

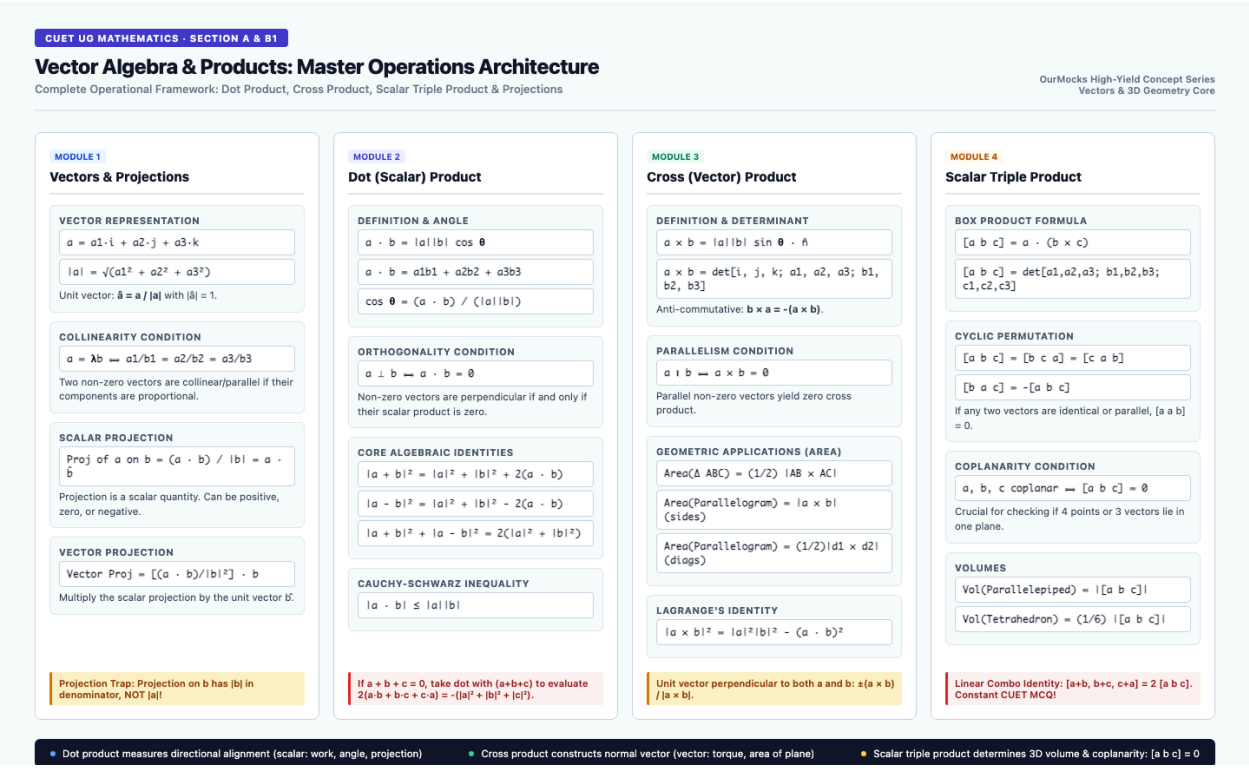


Figure 1: Figure 1: Vector Algebra, Dot vs Cross Products, Geometric Projections, and Lagrange's Identity

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Scalar Triple Product (Box Product)

The **scalar triple product** (STP) of three vectors $(a), (b), (c)$ is defined as the dot product of (a) with the cross product of (b) and (c):

$$[(a)\ (b)\ (c)] = (a) \cdot ((b) \times (c))$$

Cartesian Determinant Representation

If $(a) = a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}$, $(b) = b_1\hat{\imath} + b_2\hat{\jmath} + b_3\hat{k}$, and $(c) = c_1\hat{\imath} + c_2\hat{\jmath} + c_3\hat{k}$, then:

$$[(a)\ (b)\ (c)] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Essential Properties of STP

1. **Cyclic Permutations Preserve Value:**
$$[(a)\ (b)\ (c)] = [(b)\ (c)\ (a)] = [(c)\ (a)\ (b)]$$
2. **Non-Cyclic Permutations Flip Sign:**
$$[(b)\ (a)\ (c)] = [(a)\ (c)\ (b)] = [(c)\ (b)\ (a)] = -[(a)\ (b)\ (c)]$$
3. **Interchange of Dot and Cross:** The positions of dot and cross can be interchanged without altering the result:
$$(a) \cdot ((b) \times (c)) = ((a) \times (b)) \cdot (c)$$
4. **Repeated Vectors Result in Zero:** If any two vectors in a scalar triple product are equal (or parallel), the value is zero:
$$[(a)\ (a)\ (b)] = 0, \quad [(a)\ (b)\ (b)] = 0$$
5. **Linearity:** $[(a) + (d)\ (b)\ (c)] = [(a)\ (b)\ (c)] + [(d)\ (b)\ (c)]$, and $[\lambda(a)\ (b)\ (c)] = \lambda[(a)\ (b)\ (c)]$.

GEOMETRIC MEANING & COPLANARITY CRITERION

- **Volume of Parallelepiped:** The volume V of a parallelepiped with coterminous edges represented by $(a), (b), (c)$ is:
$$V = |[(a)\ (b)\ (c)]|$$
- **Volume of Tetrahedron:** The volume of a tetrahedron formed by coterminous edges $(a), (b), (c)$ is:
$$V_{\text{tet}} = \frac{1}{6}\,|[(a)\ (b)\ (c)]|$$
- **Condition for Coplanarity:** Three vectors $(a), (b), (c)$ are **coplanar** if and only if their scalar triple product is zero:
$$[(a)\ (b)\ (c)] = 0 \iff \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$
- **Coplanarity of 4 Points:** Four points $A((a)), B((b)), C((c)), D((d))$ are coplanar if and only if the three vectors $\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}$ are coplanar:
$$\left[\overrightarrow{AB}\ \overrightarrow{AC}\ \overrightarrow{AD}\right] = 0$$

HIGH-FREQUENCY STP IDENTITIES FOR CUET

- $[(a)+(b)\ (b)+(c)\ (c)+(a)] = 2[(a)\ (b)\ (c)]$
- $[(a)-(b)\ (b)-(c)\ (c)-(a)] = 0$ (vectors are always coplanar)
- $[(a)\times(b)\ (b)\times(c)\ (c)\times(a)] = [(a)\ (b)\ (c)]^2$

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Three-Dimensional Geometry

Direction Angles, Direction Cosines (DCs) & Direction Ratios (DRs)

Let a directed line L in 3D space make angles α, β, γ with the positive directions of the x, y, z axes respectively ($0 \le \alpha, \beta, \gamma \le \pi$). These angles are called **direction angles**.

Concept	Definition, Notation & Governing Identities
Direction Cosines (DCs)	The cosines of direction angles: $l = \cos\alpha, \quad m = \cos\beta, \quad n = \cos\gamma$.
Fundamental DC Identity	$l^2 + m^2 + n^2 = \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$.
Sine Squared Identity	$\sin^2\alpha + \sin^2\beta + \sin^2\gamma = (1-\cos^2\alpha) + (1-\cos^2\beta) + (1-\cos^2\gamma) = 3 - 1 = 2$.
Direction Ratios (DRs)	Any three real numbers a, b, c proportional to the direction cosines l, m, n: $\frac{l}{a} = \frac{m}{b} = \frac{n}{c} = \pm\frac{1}{\sqrt{a^2+b^2+c^2}}$.
Converting DRs to DCs	$l = \frac{a}{\pm\sqrt{a^2+b^2+c^2}}, \quad m = \frac{b}{\pm\sqrt{a^2+b^2+c^2}}, \quad n = \frac{c}{\pm\sqrt{a^2+b^2+c^2}}$.
DCs of Coordinate Axes	x-axis: $(1,0,0)$; y-axis: $(0,1,0)$; z-axis: $(0,0,1)$.
DRs & DCs of Segment P_1P_2	For $P_1(x_1,y_1,z_1)$ and $P_2(x_2,y_2,z_2)$: DRs are $(x_2-x_1, y_2-y_1, z_2-z_1)$; DCs are $\frac{x_2-x_1}{\lvert P_1P_2\rvert}, \frac{y_2-y_1}{\lvert P_1P_2\rvert}, \frac{z_2-z_1}{\lvert P_1P_2\rvert}$ where $\lvert P_1P_2\rvert = \sqrt{\sum (x_2-x_1)^2}$.

CRUCIAL CONCEPTUAL DIFFERENCE: DCS VS DRS

- **Uniqueness:** Direction cosines of a line are unique up to a global sign $(\pm l, \pm m, \pm n)$. Direction ratios are **infinitely many** (any non-zero scalar multiple is valid).
- **Sum of Squares:** For DCs, $l^2 + m^2 + n^2$ is **strictly equal to 1**. For DRs, $a^2 + b^2 + c^2$ can be any positive real number!
- **Angle with axes:** If a line makes equal angles with coordinate axes, $\cos^2\alpha + \cos^2\alpha + \cos^2\alpha = 1 \implies 3\cos^2\alpha = 1 \implies \cos\alpha = \pm\frac{1}{\sqrt{3}}$. The DCs are $\left(\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}\right)$.

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Equations of Lines in Space

A straight line in three-dimensional space is uniquely determined if:

1. It passes through a given point and has a given direction vector, OR
2. It passes through two specified distinct points.

Configuration	Vector Formulation	Cartesian Formulation
Point $A((a))$ & Parallel to (b)	$(r) = (a) + \lambda(b)$ where $(b) = a\hat{i} + b\hat{j} + c\hat{k}$	$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} = \lambda$ $P(x_1 + a\lambda, y_1 + b\lambda, z_1 + c\lambda)$
Two Points $A((a))$ and $B((b))$	$(r) = (a) + \lambda((b) - (a))$	$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1} = \lambda$

Angle Between Two Lines

Let line L_1 have direction vector $(b)_1 = a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$ and line L_2 have direction vector $(b)_2 = a_2\hat{i} + b_2\hat{j} + c_2\hat{k}$:

$$\cos\theta = \frac{|(b)_1 \cdot (b)_2|}{|(b)_1||(b)_2|} = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

- **Perpendicular Lines ($\theta = 90^\circ$):**

$$(b)_1 \cdot (b)_2 = 0 \iff a_1a_2 + b_1b_2 + c_1c_2 = 0 \quad (\text{or } l_1l_2 + m_1m_2 + n_1n_2 = 0)$$

- **Parallel Lines ($\theta = 0^\circ$ or 180°):**

$$(b)_1 \times (b)_2 = (0) \iff \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

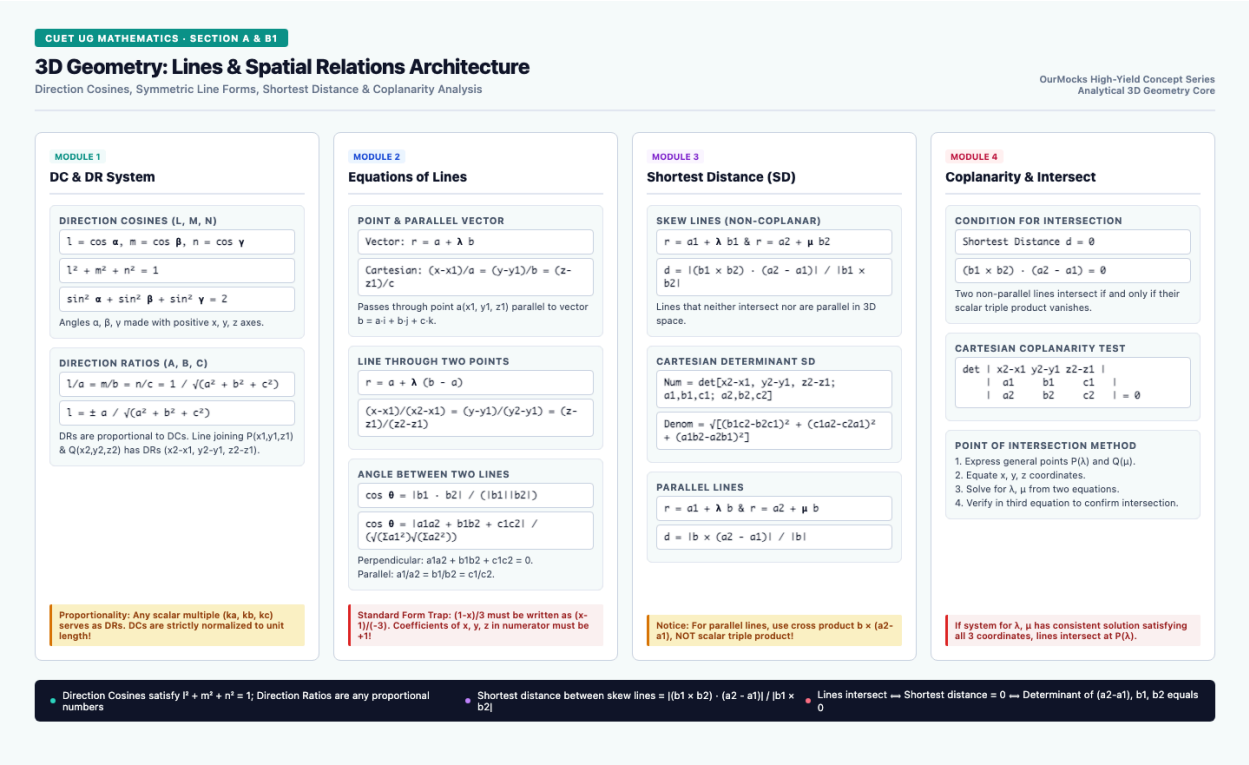


Figure 2: Figure 2: 3D Lines Architecture, Direction Angles, Skew Lines Shortest Distance vs Parallel Lines

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Shortest Distance Between Lines

In three-dimensional space, two lines can be:

1. **Intersecting:** They lie in the same plane and meet at a single point ($d = 0$).
2. **Parallel:** They lie in the same plane and never meet (distance is constant everywhere).
3. **Skew Lines:** Non-intersecting and non-parallel lines that lie in different, non-parallel planes.

Shortest Distance Between Skew Lines

Let two skew lines be given by:

$$L_1 : (r) = (a)_1 + \lambda(b)_1 \quad \text{and} \quad L_2 : (r) = (a)_2 + \mu(b)_2$$

The vector $(b)_1 \times (b)_2$ is perpendicular to both lines, representing the direction of the common perpendicular. The shortest distance d is the projection of $\left((a)_2 - (a)_1\right)$ along this common perpendicular unit normal:

SHORTEST DISTANCE FORMULAS FOR SKEW LINES

Vector Formula:

$$d = \frac{\left|\left((a)_2 - (a)_1\right) \cdot \left((b)_1 \times (b)_2\right)\right|}{\left|(b)_1 \times (b)_2\right|}$$

Cartesian Form: If $L_1 : \frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$ and $L_2 : \frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$:

$$d = \frac{\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\sqrt{(b_1c_2 - b_2c_1)^2 + (c_1a_2 - c_2a_1)^2 + (a_1b_2 - a_2b_1)^2}}$$

CRITERION FOR INTERSECTING (COPLANAR) LINES

Two lines L_1 and L_2 intersect if and only if the shortest distance between them is zero ($d = 0$):

$$\left((a)_2 - (a)_1\right) \cdot \left((b)_1 \times (b)_2\right) = 0 \iff \begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

Shortest Distance Between Parallel Lines

If two lines L_1 and L_2 are parallel, their direction vectors are identical or scalar multiples ($(b)_1 = (b)$ and $(b)_2 = (b)$):

$$L_1 : (r) = (a)_1 + \lambda(b) \quad \text{and} \quad L_2 : (r) = (a)_2 + \mu(b)$$

Here, the common perpendicular cannot be obtained from $(b) \times (b) = (0)$. Instead, taking the cross product of $\left((a)_2 - (a)_1\right)$ with (b):

$$d = \frac{\left|\left((a)_2 - (a)_1\right) \times (b)\right|}{|(b)|}$$

CRITICAL DISTINCTION: SKEW VS PARALLEL DISTANCE FORMULAS

- **Numerator for Skew Lines:** A scalar triple product (dot product with cross product), resulting in an absolute scalar: $\left|\left((a)_2 - (a)_1\right) \cdot \left((b)_1 \times (b)_2\right)\right|$.
- **Numerator for Parallel Lines:** A single vector cross product, followed by finding the magnitude: $\left|\left((a)_2 - (a)_1\right) \times (b)\right|$.
- **Mistake:** Using the skew formula on parallel lines gives $\frac{0}{0}$ (indeterminate form) because $(b) \times (b) = (0)$. Always check if direction ratios are proportional first!

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Planes in Space (Syllabus Connections)

Although planes have received varied emphasis in revised boards, CUET retains core conceptual questions involving planes, line-plane intersections, and normal vectors.

Plane Equation Form	Vector & Cartesian Form
Normal Form	$(r) \cdot \hat{n} = d$ where $\hat{n}$ is unit normal from origin, and d is perpendicular distance from origin. Cartesian: $lx + my + nz = d$.
Through Point $A((a))$ with Normal (n)	$((r) - (a)) \cdot (n) = 0$. Cartesian: $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$.
General First Degree Equation	$Ax + By + Cz + D = 0$ where (A, B, C) are direction ratios of the normal to the plane.
Intercept Form	$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ where a, b, c are non-zero intercepts on x, y, z axes.
Perpendicular Distance from Point	From $P(x_1, y_1, z_1)$ to $Ax + By + Cz + D = 0$: $d = \frac{\lvert Ax_1 + By_1 + Cz_1 + D\rvert}{\sqrt{A^2 + B^2 + C^2}}$.
Angle Between Line & Plane	Angle θ between line $(r) = (a) + \lambda(b)$ and plane $(r) \cdot (n) = d$: $\sin\theta = \frac{\lvert(b) \cdot (n)\rvert}{\lvert(b)\rvert\,\lvert(n)\rvert}$. (Note sine function because $(n) \perp$ plane).
Parallelism & Perpendicularity	Line is parallel to plane if $(b) \cdot (n) = 0$. Line is perpendicular to plane if $(b) \parallel (n)$ ($\frac{a}{A} = \frac{b}{B} = \frac{c}{C}$).

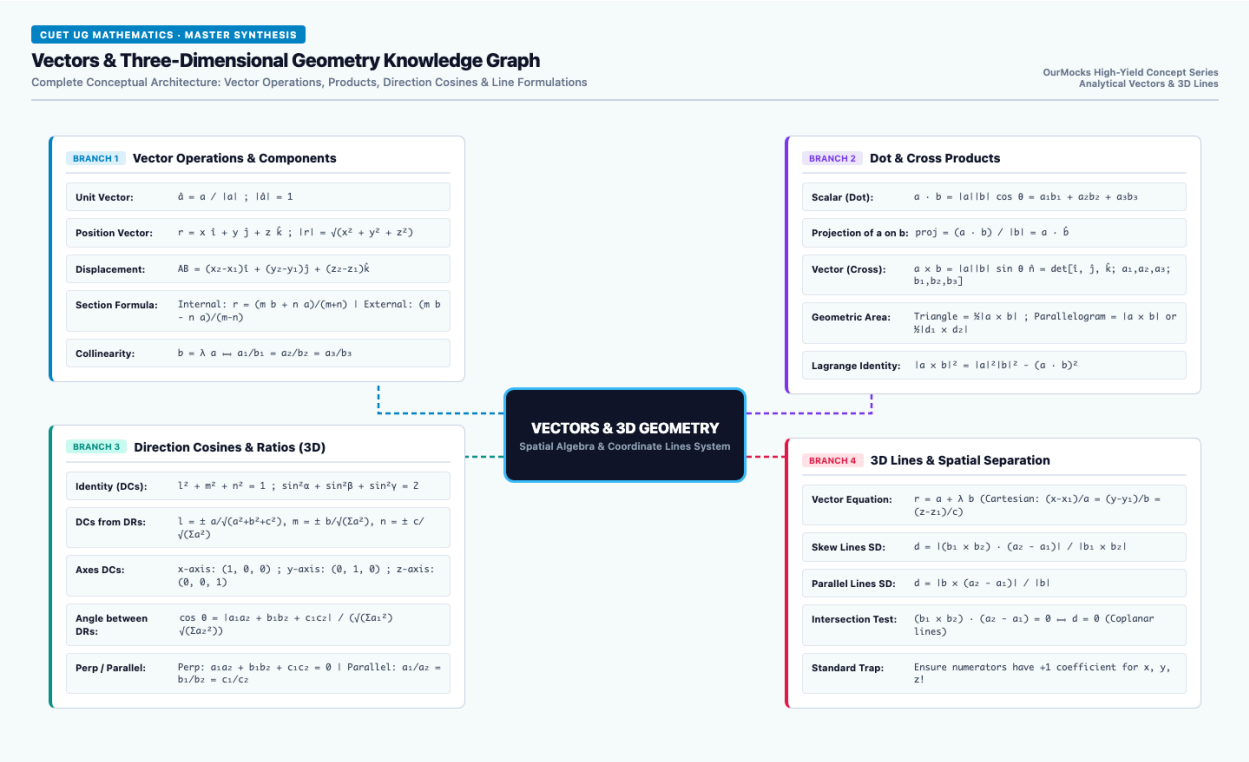


Figure 3: Figure 3: Vectors and 3D Geometry Complete Structural Synthesis Mindmap

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Common Student Mistakes & Exam Pitfalls

Pitfall Area	Incorrect Student Practice	Correct Mathematical Rule
Direction Ratios Extraction	Assuming denominator numbers are DRs without verifying numerator coefficients: $\frac{1-x}{2} \Longrightarrow a = 2$.	Rewrite $\frac{-(x-1)}{2} = \frac{x-1}{-2}$, making $a = -2$. Numerator variables must have coefficient $+1$.
Angle Between Line & Plane	Using $\cos\theta = \frac{(b)\cdot(n)}{\lvert(b)\rvert\,\lvert(n)\rvert}$ as if finding angle between two lines.	Use $\sin\theta = \frac{\lvert(b)\cdot(n)\rvert}{\lvert(b)\rvert\,\lvert(n)\rvert}$ because (n) is perpendicular to the plane.
Parallelogram Area	Multiplying diagonal cross product directly: $\lvert(d)_1 \times (d)_2\rvert$.	Area from diagonals is $\frac{1}{2}\,\lvert(d)_1 \times (d)_2\rvert$. Half factor is mandatory!
Collinearity vs Coplanarity	Using cross product for 3 vectors: $(a) \times (b) \times (c) = 0$.	Collinearity of 2 vectors: $(a) \times (b) = (0)$. Coplanarity of 3 vectors: $[(a)\ (b)\ (c)] = 0$.
Shortest Distance Denominator	Evaluating $(b)_1 \times (b)_2$ but forgetting to take its magnitude in denominator.	The denominator is the Euclidean scalar norm $\lvert(b)_1 \times (b)_2\rvert = \sqrt{x^2+y^2+z^2}$.
Direction Cosines Squares	Equating $a^2+b^2+c^2 = 1$ for arbitrary direction ratios.	Only normalized direction cosines satisfy $l^2+m^2+n^2 = 1$. DRs satisfy $a^2+b^2+c^2 = \lambda^2$.

High-Yield Formula Cheatsheet

MASTER FORMULA CHEATSHEET — VECTORS & 3D GEOMETRY

Vector Algebra:

- Unit vector: $\hat{a} = \frac{(a)}{|(a)|}$
- Internal section: $(r) = \frac{m(b)+n(a)}{m+n}$; External: $(r) = \frac{m(b)-n(a)}{m-n}$
- Scalar product: $(a) \cdot (b) = |(a)|\ |(b)| \cos\theta = a_1b_1 + a_2b_2 + a_3b_3$
- Scalar projection: $p_s = \frac{(a)\cdot(b)}{|(b)|}$; Vector projection: $(p)_v = \left(\frac{(a)\cdot(b)}{|(b)|^2}\right)(b)$
- Vector product: $(a) \times (b) = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$
- Lagrange's identity: $|(a) \times (b)|^2 + ((a) \cdot (b))^2 = |(a)|^2\ |(b)|^2$
- Parallelogram area: $|(a) \times (b)|$ (sides), $\frac{1}{2}\ |(d)_1 \times (d)_2|$ (diagonals)
- Triangle area: $\frac{1}{2}\ |(a) \times (b)| = \frac{1}{2}\ |\overrightarrow{AB} \times \overrightarrow{AC}|$
- Scalar triple product: $[(a)\ (b)\ (c)] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$
- Parallelepiped volume: $|[(a)\ (b)\ (c)]|$; Tetrahedron volume: $\frac{1}{6}\ |[(a)\ (b)\ (c)]|$
- Coplanarity: $[(a)\ (b)\ (c)] = 0$

Three-Dimensional Geometry:

- DCs & DRs: $l^2 + m^2 + n^2 = 1$, $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$
- Line through $A((a))$ parallel to (b): $(r) = (a) + \lambda(b) \iff \frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$
- Line through two points: $\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$
- Angle between lines: $\cos\theta = \frac{|a_1a_2+b_1b_2+c_1c_2|}{\sqrt{\sum a_1^2}\sqrt{\sum a_2^2}}$; Perpendicular: $\sum a_1a_2 = 0$
- Skew lines shortest distance: $d = \frac{|((a)_2-(a)_1)\cdot((b)_1\times(b)_2)|}{|(b)_1\times(b)_2|}$
- Parallel lines shortest distance: $d = \frac{|((a)_2-(a)_1)\times(b)|}{|(b)|}$
- Distance of point from plane: $d = \frac{|Ax_1+By_1+Cz_1+D|}{\sqrt{A^2+B^2+C^2}}$
- Angle between line & plane: $\sin\theta = \frac{|(b)\cdot(n)|}{|(b)|\ |(n)|}$

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Practice Questions — Part 1 (Q1 to Q5)

Q1. If (a) and (b) are two unit vectors such that $|(a)+(b)|=\sqrt{3}$, then the value of $(2(a)-3(b))\cdot((a)-(b))$ is:

(A) $\frac{3}{2}$ **(B)** $\frac{5}{2}$
(C) $\frac{7}{2}$ **(D)** $\frac{9}{2}$

Sol. Q1. Given $|(a)|=1$, $|(b)|=1$, and $|(a)+(b)|=\sqrt{3}$. Squaring both sides: $|(a)+(b)|^2=1+1+2((a)\cdot(b))=3\Longrightarrow(a)\cdot(b)=\frac{1}{2}$. Now expand: $(2(a)-3(b))\cdot((a)-(b))=2\,|(a)|^2-5((a)\cdot(b))+3\,|(b)|^2=2(1)-5\left(\frac{1}{2}\right)+3(1)=5-2.5=\frac{5}{2}$. **Answer: (B)**

Q2. If $|(a)|=4, |(b)|=2$ and the angle between (a) and (b) is $\frac{\pi}{6}$, then the value of $|(a)\times(b)|^2$ is:

(A) 16 **(B)** 48
(C) 64 **(D)** 12

Sol. Q2. By definition, $|(a)\times(b)|=|(a)|\,|(b)|\sin\theta=4\times2\times\sin\left(\frac{\pi}{6}\right)=8\times\left(\frac{1}{2}\right)=4$. Squaring gives $|(a)\times(b)|^2=4^2=16$. Alternatively, by Lagrange's Identity: $|(a)\times(b)|^2=|(a)|^2\,|(b)|^2-((a)\cdot(b))^2=16\times4-\left(4\times2\times\cos\left(\frac{\pi}{6}\right)\right)^2=64-48=16$. **Answer: (A)**

Q3. The area of a parallelogram whose diagonals are determined by the vectors $(d)_1=3\hat{\imath}+\hat{\jmath}-2\hat{k}$ and $(d)_2=\hat{\imath}-3\hat{\jmath}+4\hat{k}$ is:

(A) $5\sqrt{3}$ sq units **(B)** $10\sqrt{3}$ sq units
(C) $5\sqrt{6}$ sq units **(D)** 20 sq units

Sol. Q3. For diagonals $(d)_1$ and $(d)_2$, Area $=\frac{1}{2}\,|(d)_1\times(d)_2|$. Evaluating cross product: $(d)_1\times(d)_2=\begin{vmatrix}\hat{\imath}&\hat{\jmath}&\hat{k}\\3&1&-2\\1&-3&4\end{vmatrix}=-2\hat{\imath}-14\hat{\jmath}-10\hat{k}$. Magnitude: $|(d)_1\times(d)_2|=\sqrt{(-2)^2+(-14)^2+(-10)^2}=\sqrt{4+196+100}=\sqrt{300}=10\sqrt{3}$. Thus, Area $=\frac{1}{2}\left(10\sqrt{3}\right)=5\sqrt{3}$ sq units. **Answer: (A)**

Q4. If the vectors $(a)=2\hat{\imath}-\hat{\jmath}+\hat{k}$, $(b)=\hat{\imath}+2\hat{\jmath}-3\hat{k}$ and $(c)=3\hat{\imath}+\lambda\hat{\jmath}+5\hat{k}$ are coplanar, then the value of λ is:

(A) -4 **(B)** -2
(C) 4 **(D)** 2

Sol. Q4. Three vectors are coplanar if and only if their scalar triple product is zero: $[(a)\ (b)\ (c)]=0$. Expanding determinant: $\begin{vmatrix}2&-1&1\\1&2&-3\\3&\lambda&5\end{vmatrix}=2(10+3\lambda)+1(5+9)+1(\lambda-6)=20+6\lambda+14+\lambda-6=7\lambda+28=0\Longrightarrow\lambda=-4$. **Answer: (A)**

Q5. The direction cosines of a line making equal angles with the positive coordinate axes are:

(A) $(1,1,1)$ **(B)** $\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$
(C) $\left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right)$ **(D)** $\left(\frac{1}{2},\frac{1}{2},\frac{1}{2}\right)$

Sol. Q5. Let direction angles be $\alpha=\beta=\gamma$. From $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$, we have $3\cos^2\alpha=1\Longrightarrow\cos\alpha=\frac{1}{\sqrt{3}}$. Hence direction cosines along positive axes are $\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$. (Note: $(1,1,1)$ are direction ratios, not direction cosines). **Answer: (B)**

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Practice Questions — Part 2 (Q6 to Q10)

Q6. The direction ratios of the line $\frac{2x-1}{4} = \frac{3-y}{2} = \frac{2z+1}{6}$ are:

(A) $(4, 2, 6)$ **(B)** $(2, -2, 3)$
(C) $(2, 2, 3)$ **(D)** $(4, -2, 6)$

Sol. Q6. Convert to standard symmetric Cartesian form where coefficients of x, y, z are $+1$: $\frac{2\left(x-\frac{1}{2}\right)}{4} = \frac{-(y-3)}{2} = \frac{2\left(z+\frac{1}{2}\right)}{6} \Longrightarrow \frac{x-\frac{1}{2}}{2} = \frac{y-3}{-2} = \frac{z-\left(-\frac{1}{2}\right)}{3}$. The denominators give direction ratios $(2, -2, 3)$. **Answer: (B)**

Q7. The projection of the vector $(a) = 2\hat{\imath} + 3\hat{\jmath} + 2\hat{k}$ on the vector $(b) = \hat{\imath} + 2\hat{\jmath} + \hat{k}$ is:

(A) $\frac{5}{3}$ **(B)** $\frac{5}{\sqrt{6}}$
(C) $\frac{10}{\sqrt{6}}$ **(D)** $\sqrt{6}$

Sol. Q7. The projection of (a) on (b) is $p = \frac{(a)\cdot(b)}{|(b)|}$. We calculate $(a)\cdot(b) = (2)(1) + (3)(2) + (2)(1) = 2 + 6 + 2 = 10$, and $|(b)| = \sqrt{1^2 + 2^2 + 1^2} = \sqrt{6}$. Hence $p = \frac{10}{\sqrt{6}}$. **Answer: (C)**

Q8. The angle between the lines $\frac{x-2}{2} = \frac{y-1}{5} = \frac{z+3}{-3}$ and $\frac{x+2}{1} = \frac{y-4}{2} = \frac{z-5}{4}$ is:

(A) $\frac{\pi}{2}$ **(B)** $\frac{\pi}{3}$
(C) $\frac{\pi}{4}$ **(D)** $\frac{\pi}{6}$

Sol. Q8. The direction ratios are $(b)_1 = (2, 5, -3)$ and $(b)_2 = (1, 2, 4)$. Evaluating their dot product: $a_1a_2 + b_1b_2 + c_1c_2 = (2)(1) + (5)(2) + (-3)(4) = 2 + 10 - 12 = 0$. Since the dot product is zero, the lines are mutually perpendicular, so $\theta = \frac{\pi}{2}$. **Answer: (A)**

Q9. The shortest distance between the lines $(r) = \left(\hat{\imath} + 2\hat{\jmath} + \hat{k}\right) + \lambda\left(\hat{\imath} - \hat{\jmath} + \hat{k}\right)$ and $(r) = \left(2\hat{\imath} - \hat{\jmath} - \hat{k}\right) + \mu\left(2\hat{\imath} + \hat{\jmath} + 2\hat{k}\right)$ is:

(A) $\frac{3}{\sqrt{2}}$ units **(B)** $3\sqrt{2}$ units
(C) $\frac{9}{\sqrt{2}}$ units **(D)** 0 units

Sol. Q9. Here $(a)_1 = \hat{\imath} + 2\hat{\jmath} + \hat{k}$ and $(a)_2 = 2\hat{\imath} - \hat{\jmath} - \hat{k}$, so $(a)_2 - (a)_1 = \hat{\imath} - 3\hat{\jmath} - 2\hat{k}$. Direction cross product: $(b)_1 \times (b)_2 = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & -1 & 1 \\ 2 & 1 & 2 \end{vmatrix} = -3\hat{\imath} + 0\hat{\jmath} + 3\hat{k}$, with magnitude $|(b)_1 \times (b)_2| = 3\sqrt{2}$. Numerator: $\left|\left((a)_2 - (a)_1\right) \cdot \left((b)_1 \times (b)_2\right)\right| = |(1)(-3) + (-3)(0) + (-2)(3)| = |-9| = 9$. Shortest distance $d = \frac{9}{3\sqrt{2}} = \frac{3}{\sqrt{2}}$ units. **Answer: (A)**

Q10. The value of $[(a) - (b)\ (b) - (c)\ (c) - (a)]$ for any three arbitrary vectors $(a), (b), (c)$ is:

(A) $[(a)\ (b)\ (c)]$ **(B)** $2[(a)\ (b)\ (c)]$
(C) 0 **(D)** $-[(a)\ (b)\ (c)]$

Sol. Q10. Notice that the sum of the three vectors is $((a) - (b)) + ((b) - (c)) + ((c) - (a)) = (0)$. Since one vector can be expressed as a linear combination of the other two, the three vectors are linearly dependent and coplanar. The scalar triple product of any coplanar vectors is identically 0. **Answer: (C)**

Practice Questions — Part 3 (Q11 to Q15)

Q11. If a line makes angles α, β, γ with the coordinate axes, then the value of $\sin^2\alpha + \sin^2\beta + \sin^2\gamma$ is:

(A) 1 **(B)** 2
(C) 3 **(D)** 0

Sol. Q11. Direction cosines satisfy $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$. Substituting $\sin^2\theta = 1 - \cos^2\theta$: $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = (1 - \cos^2\alpha) + (1 - \cos^2\beta) + (1 - \cos^2\gamma) = 3 - (\cos^2\alpha + \cos^2\beta + \cos^2\gamma) = 3 - 1 = 2$. **Answer: (B)**

Q12. The shortest distance between the parallel lines $(r) = \left(\hat{\imath} + 2\hat{\jmath} - 4\hat{k}\right) + \lambda\left(2\hat{\imath} + 3\hat{\jmath} + 6\hat{k}\right)$ and $(r) = \left(3\hat{\imath} + 3\hat{\jmath} - 5\hat{k}\right) + \mu\left(2\hat{\imath} + 3\hat{\jmath} + 6\hat{k}\right)$ is:

(A) $\frac{\sqrt{293}}{7}$ **(B)** $\frac{\sqrt{293}}{49}$
(C) $\frac{1}{7}$ **(D)** $\frac{\sqrt{14}}{7}$

Sol. Q12. Here $(a)_2 - (a)_1 = 2\hat{\imath} + \hat{\jmath} - \hat{k}$ and common vector $(b) = 2\hat{\imath} + 3\hat{\jmath} + 6\hat{k}$. Cross product: $\left((a)_2 - (a)_1\right) \times (b) = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & 1 & -1 \\ 2 & 3 & 6 \end{vmatrix} = 9\hat{\imath} - 14\hat{\jmath} + 4\hat{k}$. Magnitude: $\sqrt{9^2 + (-14)^2 + 4^2} = \sqrt{81 + 196 + 16} = \sqrt{293}$. Since $|(b)| = \sqrt{4 + 9 + 36} = 7$, shortest distance $d = \frac{\sqrt{293}}{7}$. **Answer: (A)**

Q13. If $|(a)| = 3, |(b)| = 4, |(c)| = 5$ and each vector is perpendicular to the sum of the other two vectors, then $|(a) + (b) + (c)|$ is equal to:

(A) $5\sqrt{2}$ **(B)** 10
(C) 5 **(D)** $2\sqrt{5}$

Sol. Q13. Perpendicularity implies $(a) \cdot ((b) + (c)) = 0$, $(b) \cdot ((c) + (a)) = 0$, and $(c) \cdot ((a) + (b)) = 0$. Adding gives $2((a) \cdot (b) + (b) \cdot (c) + (c) \cdot (a)) = 0$. Expanding: $|(a) + (b) + (c)|^2 = |(a)|^2 + |(b)|^2 + |(c)|^2 + 2(\sum (a) \cdot (b)) = 3^2 + 4^2 + 5^2 + 0 = 9 + 16 + 25 = 50$. Hence $|(a) + (b) + (c)| = \sqrt{50} = 5\sqrt{2}$. **Answer: (A)**

Q14. The perpendicular distance of the point $P(2, 3, -5)$ from the plane $(r) \cdot \left(\hat{\imath} + 2\hat{\jmath} - 2\hat{k}\right) = 9$ is:

(A) 3 units **(B)** 6 units
(C) 9 units **(D)** 18 units

Sol. Q14. The Cartesian equation of the plane is $x + 2y - 2z - 9 = 0$. Using perpendicular distance formula: $d = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}} = \frac{|(1)(2) + (2)(3) - 2(-5) - 9|}{\sqrt{1^2 + 2^2 + (-2)^2}} = \frac{|2 + 6 + 10 - 9|}{\sqrt{9}} = \frac{9}{3} = 3$ units. **Answer: (A)**

Q15. If $\hat{a}$ and $\hat{b}$ are unit vectors and θ is the angle between them, then $\sin\left(\frac{\theta}{2}\right)$ is equal to:

(A) $\frac{1}{2}|\hat{a} - \hat{b}|$ **(B)** $\frac{1}{2}|\hat{a} + \hat{b}|$
(C) $|\hat{a} - \hat{b}|$ **(D)** $\frac{1}{2}\left(\hat{a} \cdot \hat{b}\right)$

Sol. Q15. Expanding: $|\hat{a} - \hat{b}|^2 = |\hat{a}|^2 + |\hat{b}|^2 - 2\left(\hat{a} \cdot \hat{b}\right) = 1 + 1 - 2\cos\theta = 2(1 - \cos\theta) = 4\sin^2\left(\frac{\theta}{2}\right)$. Taking positive square root $(0 \le \theta \le \pi)$: $|\hat{a} - \hat{b}| = 2\sin\left(\frac{\theta}{2}\right) \Longrightarrow \sin\left(\frac{\theta}{2}\right) = \frac{1}{2}|\hat{a} - \hat{b}|$. (Note: companion formula for cosine is $\cos\left(\frac{\theta}{2}\right) = \frac{1}{2}|\hat{a} + \hat{b}|$). **Answer: (A)**